Mikkel Paltorp

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Work Experience

Postdoctoral Researcher (SciML for Structural Health Monitoring) 2025 — Present
Technical University of Denmark (DTU Compute) *Kgs. Lyngby, Denmark*

- Developing scalable Gaussian process algorithms for structural health monitoring.
- Implementing structured matrix algorithms for Gaussian process inference in SSSMatrices.jl.

Postdoctoral Researcher (Conic Optimization) 2024
Technical University of Denmark (DTU Compute) / OPTUM CE *Kgs. Lyngby / Copenhagen, Denmark*

- Adapted the open-source conic solver Clarabel for OPTUM CE's finite element limit analysis problems.
- Collaborated with OPTUM CE on integrating the adapted solver into their engineering software.

PhD Researcher in Computational Acoustics 2020 — 2024
Technical University of Denmark (DTU Electro) *Kgs. Lyngby, Denmark*

- Applied fast multipole methods and hierarchical matrices to viscothermal acoustics, enabling simulations orders of magnitude larger than previously feasible.
- Created BoundaryIntegralEquations.jl, now used at the Technical University of Munich.
- Taught FEM/BEM in the course Numerical Acoustics and supervised student projects.

Selected Scientific Software

SSSMatrices.jl / arXiv:2610.05856 *(release pending paper acceptance) Creator*
Structured matrix factorizations, solves, and log-determinants for fast Gaussian process inference.

BoundaryIntegralEquations.jl / FMM2D.jl *Creator*
Viscothermal acoustics with boundary elements and Julia bindings to Flatiron's fast multipole library.

Pardiso.jl *Maintainer*
Julia interface to the MKL/Panua PARDISO sparse direct solvers, used throughout the Julia ecosystem.

Skills

- **Programming:** Julia, Python, MATLAB, R, C, Rust.
- **Linear algebra:** Sparse direct solvers, hierarchical/semiseparable matrices, FMM, Krylov methods.
- **Numerical methods:** FEM/BEM, mathematical programming, Gaussian processes, adjoint methods.
- **Libraries:** JuMP, CVXPY, Clarabel, PARDISO, ForwardDiff, Mooncake, marimo.
- **Tools:** Git, CI/CD, LaTeX, Typst.
- **Languages:** Danish (native), English (fluent).

Education

M.Sc.Eng in Mathematical Modelling and Computation 2020
Technical University of Denmark (DTU) *Kgs. Lyngby, Denmark*
Thesis: implemented Gaussian process algorithms with linear time and storage complexity.

B.Sc.Eng in Mathematics and Technology 2017
Technical University of Denmark (DTU) *Kgs. Lyngby, Denmark*

McGill University - Exchange semester, Montreal, Canada Fall 2016

Community

Technical blog (mipals.github.io/blog) *Author*
Posts on numerical linear algebra, optimization, and FEM, with Julia code and interactive notebooks.

Copenhagen Julia user group *Organizer*
Organizing events with around 30 participants and speakers from industry and academia.